Python Assignment

Question 1: What is Python and why is it widely used in Data Analytics? Mention any three reasons.

ANSWER : Python is a general-purpose programming language used to work with data, build software, and automate tasks. It is widely used in data analytics because:

1. It has simple, readable syntax.
2. It has useful data analysis libraries, such as pandas and NumPy.
3. It can create charts and connect to databases.

Question 2: Explain the difference between List and Tuple in Python.

ANSWER : A list and a tuple are both ordered collections of values. A list can be changed after it is created, while a tuple cannot be changed. Lists use square brackets, and tuples use parentheses.

```
my_list = [1, 2, 3]
my_tuple = (1, 2, 3)
```

Question 3: What is a function in Python? Why are functions useful?

ANSWER : A function is a named block of code that performs a specific task. Functions are useful because they help organize a program, allow code to be reused, and reduce repetition.

Question 4: Write a Python program to take a user's name as input and print a greeting message.

ANSWER :

```
 name = input("Enter your name: ")
print("Hello,", name + "!")
```

Question 5: Write a Python program to check whether a number is even or odd using conditional statements.

ANSWER :

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("The number is even.")
else:
    print("The number is odd.")
```

Question 6: Write a program to print numbers from 1 to 10 using a loop.

ANSWER :

```
for number in range(1, 11):
print(number)
```

Question 7: Create a list of five numbers and print the maximum number from the list.

ANSWER :

```
numbers = [8, 3, 12, 5, 9]
print("The maximum number is:", max(numbers))
```

Question 8: Write a Python program to remove duplicate values from a list using a set.

ANSWER :

```
 numbers = [1, 2, 2, 3, 1, 4]
unique_numbers = list(set(numbers))
print("List without duplicates:", unique_numbers)
```

Question 9: Write a function that returns the square of a number.

ANSWER :

```
def square(number):
    return number ** 2

print(square(5))
```

Question 10: Write a Python program to count how many times a number appears in a list.

ANSWER :

```
numbers = [2, 3, 4, 2, 5, 2]
target = int(input("Enter the number to count: "))

count = numbers.count(target)
print("The number appears", count, "times.")
```